Lesson 2 : Manufacturing processes

Manufacturing processes can be broadly divided into two groups: primary and secondary. The former such as forging and casting provide basic shape and size to the material. Secondary processes provide the final shape and size with tighter control on dimension and surface characteristics. This second group is performed by so called "machine tools".

Machine tools form the basis of precision manufacturing. As such they are indispensable for the production of nearly all modern technology, be it cars, planes, household appliances, computers, solar panels, wind turbines, or other machine tools. **<u>They</u>** are so fundamental to manufacturing that the United Nations Industrial Development Organization (UNIDO) defines "industrialisation" as the capacity to produce machine tools.

Machine tools are stationary operating machines that perform the geometric shaping of workpieces. They are mostly subtractive technologies, dealing with any process **<u>in which</u>** material is removed gradually from a workpiece. There is a great diversity in specialized machine tools, but common examples are sawing and cutting machines, drilling machines, grinding machines, turning machines (lathes) and milling machines.

Machine tools can process different materials, but today most of them are used for the shaping of metal parts. Milling machines have become the most common metal shaping tools in the mechanical manufacturing industry. Milling is a process of cutting away material by feeding a workpiece past a rotating cutting tool. A milling machine operates like a human sculptor. It can have grinding, cutting and polishing heads.

Lathes or turning machines are similar to milling machines but **<u>they</u>** are used to shape cylindrical workpieces. In a milling machine the workpiece remains (relatively) stationary while material is removed by moving tools heads. In a lathe, the workpiece is rotating while the tool head remains (relatively) stationary.